

EGE YESILYURT

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Education

York University — North York, ON

Expected July 2027

Bachelor of Engineering, Major in Software Engineering

Relevant Coursework: Object-Oriented Programming, Operating Systems, Data Structures and Algorithms, Software Design, Design and Analysis of Algorithms, E-Commerce Systems

Projects

z9bra — CSP Puzzle Generator & Solver | [Website](#) | [App Source](#) | [Generator](#)

Live, June 2026 • Open Source

- **Web app:** Shipped a Wordle-style daily logic puzzle now at **500+ unique players and 60 active daily players**, using date-seeded selection so everyone worldwide gets the same grid on a fixed Toronto reset, with a commit-based solving engine, **Satori** share cards and account-synced streaks.
- **Generator (backend):** Wrote the **Python** engine behind those puzzles, using **Google OR-Tools CP-SAT** as a uniqueness oracle so every export is provably **single-solution**, with a propagation tracer scoring difficulty by human deduction depth. **190+ tests**, mypy-strict.

Technologies: Next.js, React, TypeScript, Tailwind, Satori, Vercel; Python, OR-Tools CP-SAT, Pydantic v2, pytest, Hypothesis, mypy

OverTerm — Agent-Aware Floating Terminal for macOS | [Source Code](#)

v1.0.3, August 2026 • Open Source

- Built and shipped a native macOS terminal in **Rust + Tauri v2** that detects whether an AI coding agent is running, finished, or waiting, shrinking to a status bar while it works and expanding when it needs you.
- Determined agent state from a **vt100 screen model** of the pseudoterminal rather than the byte stream, so it works on any CLI tool, and read exact turn events out of Claude Code hooks as OSC markers.
- Shipped it as a **one-command install** behind a tag-triggered **GitHub Actions** pipeline that verifies every artifact before publishing and auto-bumps the Homebrew cask. **177 tests**, MIT, with outside users filing issues.

Technologies: Rust, Tauri v2, TypeScript, xterm.js, portable-pty, vt100, objc2, GitHub Actions, Homebrew

BundesPredict — Calibrated Match Predictor with an LLM Agent | [Website](#) | [Source Code](#)

Live, July 2026

- Built a Bundesliga match-outcome predictor on a **hand-written Dixon–Coles Poisson model** (weighted MLE via SciPy, time decay, promoted-team shrinkage); **backtested walk-forward over 1,800+ out-of-sample matches** to a calibrated **RPS of 0.205**, within a hundredth of the de-vigged market line.
- Designed an **LLM tool-calling agent** that maps plain-English match context into **bounded, server-clamped expected-goals adjustments**. The deterministic model owns every probability; the LLM only proposes typed, audited nudges grounded in scraped squad data.
- Architected a Dockerized **FastAPI + Next.js + Postgres** monorepo with a **230+ test suite** (parameter-recovery, calibration, invariants), **GitHub Actions CI/CD**, and a weekly refit job; deployed on Vercel + Railway + Neon.

Technologies: Python, FastAPI, NumPy/SciPy, scikit-learn, Anthropic SDK, Next.js, TypeScript, PostgreSQL, Docker, GitHub Actions

Deximon — Pokémon TCG Collector Platform | [Website](#) | [Source Code](#)

Live, Summer 2026 • 6-Person Team

- Shipped a collector platform (digital binder, card-recognition scanner, listing-scoped marketplace) as the **largest contributor on a 6-person team**, with **58 of 109 commits**.
- **Reviewed 12 teammate pull requests** with written scores and follow-up fix PRs, catching a WebSocket handler that crashed on dead sockets, an **N+1 query** on the marketplace feed, and notification delivery that broke multi-tab users.
- Built binder persistence end to end across a **Docker Compose monorepo**: Next.js web, FastAPI backend, and a FastAPI scanner service on **AWS Rekognition**. **80+ backend tests**.

Technologies: Next.js, React, TypeScript, Python, FastAPI, PostgreSQL, Docker Compose, WebSockets, AWS Rekognition, JWT, OpenAPI

Spocity — Spotify-Powered 3D Voxel City | [Website](#) | [Source Code](#)

Live, July 2026

- Built and shipped a web app that turns a user's **Spotify listening history** into an explorable **3D voxel city** rendered in-browser with **React Three Fiber**, where each artist is a building whose height maps to a decay-weighted play score, auto-zoned into 10 genre districts.
- Engineered an **exponential-decay scoring engine** (~90-day half-life) with a nightly **Celery + Redis** recompute that re-tiers buildings and emits growth/decay events, so the skyline evolves as listening habits shift.
- Integrated **Spotify OAuth 2.0** (Auth Code + PKCE) into Django session auth and built a client-side **WebGL postcard generator** for shareable snapshots; deployed frontend on **Vercel**, Django backend + Postgres on **Railway**.

Technologies: Next.js, React, TypeScript, React Three Fiber, Django REST Framework, Celery, Redis, PostgreSQL, Spotify API, OAuth 2.0, Docker

Skills

Technical: Python, TypeScript, JavaScript, Rust, Java, SQL, HTML/CSS, Django REST Framework, FastAPI, Next.js, React, React Three Fiber, Tauri, Celery, Docker, Kubernetes, PostgreSQL, MySQL, Redis, Git, GitHub Actions, REST APIs, WebSockets, OAuth 2.0

Languages: Fluent in English, Turkish; N3 Level Japanese